Abstract

The present invention discloses a system and method for multi-agent interaction sincerity audit based on logical continuity. The system comprises a **causal logic continuity verification unit**, a privacy desensitization and signature verification unit, a **Distributed Sincerity Audit Contract**, and a **Three-Tier Gradient Disciplinary State Machine**. During interactions, the system extracts four-stage feature vectors and hashes of "demand, solution, action, effect", and performs temporal causal matching verification to identify **Logical Disconnection**. In a local trusted isolation sandbox, it generates Nonce-signed desensitized Boolean proofs and transmits them via an encrypted channel. After passing signature verification and optimistic lock concurrency control, the distributed audit contract updates the **dynamic audit weight score** of Logic Actors incrementally using a sliding window model with a decay factor. The state machine updates statuses according to weight scores, and imposes penalties on fraudulent nodes including token settlement freezing, free quota control, gateway socket physical blocking, or token bucket handshake latency injection. While absolutely protecting data privacy, the present invention implements accurate identification and protocol-stack-level physical interception of malicious free-riding and payment-evading nodes.

Figure 1 is designated as the accompanying drawing for the abstract.